Chapter 10. Trees

Staying Connected

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Networks should be connected with as few links as possible. Graphs that are connected but lose connectivity when any edge is removed are the trees of this chapter.

10.1 Minimally connected graphs

Motivation. "Minimally connected", "connected without cycles", "unique paths" and "connected with $n-1$ edges" all describe the same graphs, the trees. Counting labelled trees gives Cayley's surprisingly clean n^{n-2}, proved here by Joyal's bijection with functions $[n] \to [n]$.

Walks contain paths [Supplement]. If there is a walk from x to y, there is a path: take a shortest walk; if it visited a vertex twice, cutting out the portion between the two visits would give a shorter walk.

Theorem 10.1

For a connected simple graph G the following are equivalent:
(1) G is **minimally connected**: deleting any edge disconnects it;
(2) G contains no cycle.

Definition 10.2 (Tree)

A connected simple graph satisfying (1), equivalently (2), is a **tree**.

Proof of Theorem 10.1.

Step (1)⇒(2). If G had a cycle C containing the edge ab, delete ab. A path using ab can be rerouted along the rest of C, giving a walk and hence a path. So $G-ab$ is still connected, contradicting (1).

Step (2)⇒(1). If deleting an edge AB leaves G connected, a path P from B to A avoids AB, and $AB \cup P$ is a cycle. ■

Corollary 10.3

A connected graph is a tree iff every two vertices are joined by exactly one path.

Proof.

Step (unique paths ⇒ tree). If deleting an edge rs left the graph connected, r and s would be joined both by the edge rs and by a path avoiding it.

Step (tree ⇒ unique paths) [Supplement]. Suppose two different paths P, Q go from x to y. Let u be the last vertex up to which they agree, so they leave u along different edges. Let w be the first vertex after u on P that also lies on Q after u (it exists, since y is such a vertex). The segments of P and Q from u to w meet only at u and w, so together they form a cycle, contradicting Theorem 10.1. ■

Deleting one edge of an n-cycle gives a tree with $n-1$ edges; experiments suggest this is always the number of edges.

Theorem 10.4

Every tree on n vertices has $n-1$ edges. Conversely, every connected graph on n vertices with $n-1$ edges is a tree.

Lemma 10.5

A tree on $n \geq 2$ vertices has at least two vertices of degree 1 (**leaves**).

Proof. Take a path p of maximum length; it has two distinct endpoints. If an endpoint a had another neighbour b, then either $b \notin p$ and p could be extended, or $b \in p$ and p together with ab would contain a cycle. Both are impossible, so both endpoints are leaves. ■

Proof of Theorem 10.4.

Step 1 (trees). Induction on n; $n = 1$ is trivial. A tree T on $n+1$ vertices has a leaf l (Lemma 10.5). $T - l$ is connected and acyclic, hence a tree on n vertices with $n - 1$ edges, so T has n edges.

Step 2 (converse) [Supplement]. Let G be connected with n vertices and $n - 1$ edges. While G has a cycle, delete an edge of a cycle; connectivity is preserved (proof of Theorem 10.1). This ends with a tree on the same n vertices, which by Step 1 has $n - 1$ edges. So no edge was deleted, and G was a tree. ■

A **forest** is a graph whose components are trees.

Proposition 10.6

A forest on n vertices with k components has $n - k$ edges.

Proof. Apply Theorem 10.4 to each component: each has one more vertex than edges. ■

Counting labelled trees. Unlabelled trees are counted in Chapter 18 of the book. Trees on the vertex set $[n]$ number $1, 1, 3, 16$ for $n = 1, 2, 3, 4$, suggesting n^{n-2}; this time the "law of small numbers" does not mislead.

Theorem 10.7 (Cayley's formula)

There are $A_n = n^{n-2}$ trees with vertex set $[n]$.

Proof (Joyal).

Step 1 (doubly rooted trees). In each tree mark a vertex Start and a vertex End (possibly equal), in n^2 ways. There are $n^2 A_n$ such *doubly rooted trees*. We give a bijection from the n^n functions $f : [n] \to [n]$ to them, so $n^2 A_n = n^n$.

Step 2 (functions to trees). Let $C = \{c_1 < \cdots < c_k\}$ be the elements lying on cycles of f ($f^i(x) = x$ for some $i \geq 1$), and $d_i := f(c_i)$. Write $d_1, d_2, \ldots, d_k$ along a path, marking d_1 as Start and d_k as End. For $j \notin C$ add the edge $\{j, f(j)\}$. The result is connected, since iterating f from any j reaches C. It has $n - 1$ edges ($k - 1$ on the path, one for each $j \notin C$), so it is a tree by Theorem 10.4.

Step 3 (inverse). Given a doubly rooted tree, for j off the Start–End path let $f(j)$ be the neighbour of j on its unique path to the Start–End path. On the path, list the vertices from Start to End as $d_1, \ldots, d_k$, sort them as $c_1 < \cdots < c_k$, and set $f(c_i) := d_i$. This recovers f uniquely, so the map is a bijection. ■

Example 10.8

$n = 8$: $f(1) = 3$, $f(2) = 4$, $f(3) = 1$, $f(4) = 5$, $f(5) = 5$, $f(6) = 7$, $f(7) = 8$, $f(8) = 6$. The cycles are $(13), (5), (678)$, so $C = \{1, 3, 5, 6, 7, 8\}$ and $(d_i) = (3, 1, 5, 7, 8, 6)$. Hence the Start–End path is $3 - 1 - 5 - 7 - 8 - 6$, and $2 - 4 - 5$ hangs off it.

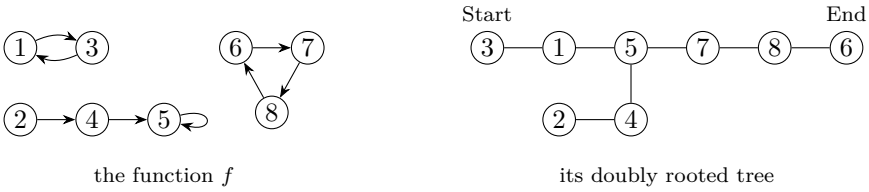


Figure. *Joyal's bijection for Example 10.8: the cyclic part of f becomes the Start–End path (read $f(c_1), \dots, f(c_k)$ with $c_1 < \cdots < c_k$); the other elements hang off it along $j \mapsto f(j)$.*

A **rooted tree** has one marked vertex (the root); there are n^{n-1} rooted trees on $[n]$. A **rooted forest** is a forest whose components are rooted trees.

Corollary 10.9

There are $(n+1)^{n-1}$ rooted forests on $[n]$.

Proof. Joining every root to a new vertex $n+1$ turns a rooted forest on $[n]$ into a tree on $[n+1]$. Conversely, from a tree on $[n+1]$, delete $n+1$ and declare its former neighbours roots. These maps are inverse, and Theorem 10.7 counts $(n+1)^{n-1}$ trees on $[n+1]$. ■

10.2 Minimum-weight spanning trees. Kruskal's greedy algorithm

Motivation. Among the n^{n-2} spanning trees of a weighted K_n we want the cheapest, and trying them all is hopeless. Greedily taking the cheapest edge that creates no cycle is optimal, thanks to an exchange property of forests (the matroid property).

A **spanning tree** of a connected G is a tree with the same vertex set using edges of G. Every connected graph has one: delete edges of cycles until none remain. K_n has n^{n-2} spanning trees.

Example 10.10 (Minimum-weight spanning tree)

Given a connected simple graph G and weights $w : E(G) \to \mathbb{R}_+$, find a spanning tree T minimising $w(T) = \sum_{e \in T} w(e)$. (For $n = 20$ cities there are $20^{18} > 2.5 \cdot 10^{23}$ spanning trees; at 10^9 per second, checking them all would take more than 91 years.)

Kruskal's greedy algorithm. Start with $T = \emptyset$. At step i, add an edge $e_i \notin T$ of minimum weight among those whose addition creates no cycle. T is always a forest. While it has fewer than $n-1$ edges it is disconnected, and since G is connected some edge of G joins two components of T. So the algorithm runs $n-1$ steps and ends with a spanning tree.

Greedy methods are not always correct: for two disjoint edges of minimum total weight in a path with weights $2, 1, 2$ plus a disjoint edge of weight 10, greedy picks $1 + 10 = 11$ instead of $2 + 2 = 4$ (minimum-weight matching, Chapter 11).

Lemma 10.11 (Exchange property)

If F, F' are forests on the same vertex set and F has fewer edges than F', then some edge e of F' can be added to F so that $F \cup e$ is still a forest.

Proof. Otherwise every edge of F' joins two vertices of the same component of F. Then each component of F' lies inside a component of F, so F' has at least as many components as F. By Proposition 10.6 it has at most as many edges, a contradiction. ■

Theorem 10.12

Kruskal's greedy algorithm finds a minimum-weight spanning tree.

Proof.

Step 1. Let $T = \{t_1, \dots, t_{n-1}\}$ be the greedy tree and suppose a spanning tree $H = \{h_1, \dots, h_{n-1}\}$ is lighter, both listed by increasing weight. Note $w(t_1) \le \cdots \le w(t_{n-1})$, since the set of eligible edges only shrinks.

Step 2. Let i be the smallest index with $\sum_{j \le i} w(h_j) < \sum_{j \le i} w(t_j)$; it exists because H is lighter overall. Since the inequality fails for $i-1$, we get $w(h_i) < w(t_i)$.

Step 3. Apply Lemma 10.11 to $T_{i-1} = \{t_1, \dots, t_{i-1}\}$ and $H_i = \{h_1, \dots, h_i\}$: some h_j, $j \le i$, can be added to T_{i-1} without a cycle. As $w(h_j) \le w(h_i) < w(t_i)$, the greedy algorithm would not have chosen t_i at step i, a contradiction. ∎

10.3 Adjacency matrices of graphs

Motivation. Encoding a graph by a matrix turns walks into matrix powers, so linear algebra answers counting questions: walk counts, connectivity and, in the next section, spanning trees.

Definition 10.13 (Adjacency matrix)

For a graph G on labelled vertices $1, \dots, n$, the **adjacency matrix** $A = A_G$ has A_{ij} equal to the number of edges between i and j (from i to j for digraphs). It is symmetric for undirected graphs.

Example 10.14

For the undirected graph on A, B, C, D with edges AB, AC, AD, BD:

$$A_G = \begin{pmatrix} 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 \end{pmatrix}.$$

Example 10.15

For the digraph with edges $A \to B$, $A \to C$, $B \to D$, $D \to A$:

$$A_H = \begin{pmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 \end{pmatrix}.$$

Theorem 10.16

$(A^k)_{ij}$ is the number of walks of length k from i to j.

Proof. Induction on k; for $k = 1$ a walk of length 1 is an edge. If b_{iz} is the number of walks of length k from i to z (by induction $B = A^k$), then the walks of length $k+1$ from i to j with next-to-last vertex z number $b_{iz}a_{zj}$. Summing over z gives $(BA)_{ij} = (A^{k+1})_{ij}$. ∎

Theorem 10.17

A simple graph G on n vertices is connected iff all entries of $(I+A)^{n-1}$ are positive.

Proof. Walks contain paths, and paths have length at most $n-1$. So G is connected iff for all $i \neq j$ some $(A^k)_{ij} > 0$ with $1 \le k \le n-1$; the diagonal entries are always positive because of the term I. Since $(I+A)^{n-1} = \sum_{k=0}^{n-1} \binom{n-1}{k} A^k$ with positive coefficients and nonnegative matrices, the claim follows. ∎

10.4 The number of spanning trees of a graph

Motivation. Kirchhoff's matrix-tree theorem counts spanning trees as a determinant. Square submatrices of an incidence matrix have determinant ± 1 exactly on spanning trees, and the Cauchy–Binet formula adds these up.

A **spanning tree** of a digraph is a subgraph whose underlying undirected graph is a spanning tree.

Definition 10.18 (Incidence matrix)

For a loopless digraph with vertices $v_1, \dots, v_n$ and edges $e_1, \dots, e_m$, the **incidence matrix** A is $n \times m$ with $a_{ij} = 1$ if v_i is the start of e_j, -1 if v_i is its end, and 0 otherwise.

Theorem 10.19

Let A be the incidence matrix of a loopless digraph G and A_0 the matrix obtained by deleting any one row. Then the number of spanning trees of G is $\det(A_0 A_0^{\top})$.

Proof. Let the last row be deleted. If $m < n-1$, G is disconnected and both sides are 0.

Step 1 (claim). For every $(n-1) \times (n-1)$ submatrix B of A_0 (a choice of $n-1$ columns, i.e. edges, forming a subgraph G'), $|\det B| = 1$ if G' is a spanning tree and $\det B = 0$ otherwise. We use induction on n; for $n = 2$, B is a 1×1 matrix ± 1 for the single edge.

Step 2 (a leaf other than v_n). If some v_i, $i \neq n$, has degree 1 in G', its row in B has a single nonzero entry ± 1. Expanding along it reduces to the same situation for $G - v_i$ and $G' - v_i$, and G' is a spanning tree iff $G' - v_i$ is one. Induction applies.

Step 3 (no such leaf). Then G' is not a spanning tree (a tree has two leaves, Lemma 10.5). Having $n-1$ edges, it has a vertex of degree 0: otherwise all degrees except possibly that of v_n would be at least 2, giving at least $\frac{2(n-1)+1}{2} > n-1$ edges. If that vertex is v_i, $i \neq n$, then B has a zero row. If it is v_n, each column of B contains one 1 and one -1, so the rows of B sum to 0. Either way $\det B = 0$.

Step 4 (Cauchy–Binet). $\det(A_0 A_0^{\top}) = \sum_B (\det B)^2$, the sum over all $(n-1) \times (n-1)$ submatrices B of A_0. By Step 1 this is the number of spanning trees. ∎

Theorem 10.20 (Matrix-tree theorem)

Let U be a simple undirected graph with vertices $v_1, \dots, v_n$ and L_0 the $(n-1) \times (n-1)$ matrix with $\ell_{ii} = \deg v_i$, $\ell_{ij} = -1$ if $v_i v_j$ is an edge, and 0 otherwise ($1 \le i, j \le n-1$). Then U has exactly $\det L_0$ spanning trees.

Proof.

Step 1. Replace each edge of U by two opposite directed edges to get G, and let A_0 be its incidence matrix without the last row. Entry (i,i) of $A_0 A_0^{\top}$ counts the edges of G at v_i, which is $2 \deg_U v_i$.

Entry (i,j), $i \neq j$, gets -1 from each of the two edges between v_i and v_j if they are adjacent. So $A_0 A_0^{\top} = 2L_0$ and $\det(A_0 A_0^{\top}) = 2^{n-1} \det L_0$.

Step 2. Each spanning tree of U yields 2^{n-1} spanning trees of G (orient each of its $n-1$ edges in either direction), and every spanning tree of G arises this way. By Theorem 10.19, $2^{n-1}\tau(U) = 2^{n-1} \det L_0$. ∎

Example 10.21

K_n has n^{n-2} spanning trees. Here L_0 has $n-1$ on the diagonal and -1 elsewhere. Adding all rows to the first gives a first row $(1,\dots,1)$; adding it to every other row gives an upper triangular matrix with diagonal $1, n, \dots, n$. So $\det L_0 = n^{n-2}$.

Example 10.22

The **complete bipartite graph** $K_{m,n}$ (parts of sizes m and n, all mn edges between them) has $n^{m-1}m^{n-1}$ spanning trees. Delete a vertex of the n-part. L_0 has the diagonal blocks nI_m and mI_{n-1}, with -1 entries between the parts. Adding all rows to the first gives $(1,\dots,1,0,\dots,0)$, and adding that row to the last $n-1$ rows makes the matrix block triangular, with determinant $n^{m-1}m^{n-1}$.

Theorem 10.23 (Matrix-tree theorem, eigenvalue version)

Let L be the $n \times n$ **Laplacian** of U (defined like L_0 but over all n vertices), with eigenvalues $\lambda_1, \dots, \lambda_{n-1}, \lambda_n = 0$. Then the number of spanning trees of U is $\frac{1}{n}\lambda_1\lambda_2\cdots\lambda_{n-1}$.

Proof [Supplement]. The book omits the proof. The rows of L sum to 0, so 0 is an eigenvalue. Write $\det(xI - L) = x\prod_{i<n}(x-\lambda_i)$; the coefficient of x is $(-1)^{n-1}\lambda_1\cdots\lambda_{n-1}$. On the other hand, expanding the determinant, the coefficient of x in $\det(xI-L)$ is $(-1)^{n-1}$ times the sum of the n principal $(n-1)\times(n-1)$ minors of L. Deleting row and column k from L gives L_0 for the labelling with v_k last, so by Theorem 10.20 every such minor equals the number $\tau(U)$ of spanning trees. Hence $\lambda_1\cdots\lambda_{n-1} = n\,\tau(U)$. ∎

If U is d**-regular** (all degrees d), then $L = dI - A$, so the eigenvalues of L are $d - \alpha_i$ for the eigenvalues α_i of A.

Example 10.24

For K_n, $A + I = J$ (all ones) has rank 1 and trace n, so its eigenvalues are $n, 0, \dots, 0$. Those of A are $n-1, -1, \dots, -1$, those of $L = (n-1)I - A$ are $0, n, \dots, n$, and Theorem 10.23 gives $\frac{1}{n}n^{n-1} = n^{n-2}$ again.